OcuDetect: Multi-label Classification of Ocular Diseases Progress Report

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https://github.com/selinlyx/OcuDetect/tree/main

Project Description

Vision impairment impacts 2.2 billion people around the world, and is largely attributable to ocular diseases [1]. Early detection of these diseases is critical to preventing irreversible vision loss [2], yet referral misdiagnosis occurs in 49% of ophthalmologic cases, leading to harm in 26% of patients [3]. To exacerbate this issue, analyzing retinal images from clinical modalities such as colour fundus photography (CFP) is time-consuming and requires specialists who are not always accessible [4, 5].

Deep learning (DL) models show promise in addressing accuracy and efficiency challenges in traditional diagnosis workflows. Convolutional neural networks (CNNs) are well-suited for this problem, as they can effectively extract spatial features from retinal images to output multi-label disease predictions. More recently, vision transformers (ViTs) have demonstrated superior performance to CNNs in detecting certain ocular pathologies by leveraging self-attention to capture global feature relationships across the retinal image [6]. Despite the progress, broad-spectrum classification can remain a challenge [4], and the sensitive, high-stakes nature of deploying such tools in clinical settings motivates ongoing efforts to enhance diagnostic accuracy. This report details the current developments of OcuDetect, a model for broad-spectrum multi-label ocular disease classfiication from 8 classes of CFP images (Figure1), with the goal of improving diagnostic accuracy and accessibility.

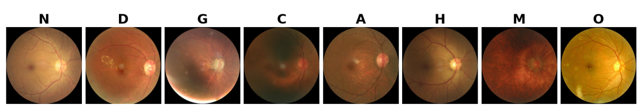

Figure 1: 8 classes of the ODIR-5K dataset. N - normal, D - diabetic retinopathy, G - glaucoma, C - cataract, A - age-related macular degeneration, H - hypertension, M - myopia, O - other.

Data Processing

OcuDetect has been trained on the ODIR-5K dataset, which contains 8 classes and multi-label annotations [7]. The following preprocessing steps have been performed on the data. Statistics on the cleaned data are shown in Table1 and Table2, and a sample preprocessed image is shown in Figure4.

1. **Cropping to the region of interest:** CFPs have circular field of views; the background has no diagnostic value and can lead to false positives [8]. Thus, background pixels were blackened using the Fundus Circle Cropping (FCC) tool by Berens Lab [9].
2. **Image rescaling:** images were rescaled using the FCC tool to 224x224 to achieve uniformity in pixel size and compliance with the fixed input size of the backbones.
3. **Contrast enhancement:** CFPs can have low contrast, obscuring important diagnostic features like blood vessels [10]. Thus, the Contrast Limited Adaptive Histogram Equalization tool by OpenCV was used to improve contrast.
4. **Data annotation cleaning:** the annotation file was cleaned to remove images flagged with keywords that indicate poor quality. Patient-level labels (unioned from both eyes) were separated into image-level labels using image-specific diagnostic keywords.

5. **Data splitting:** iterative stratification was used to split the data into 70% train, 15% validation, and 15% test.

Table 1: Proportions of 8 classes across the train, validation, and test sets after cleaning.

Disease	Train	Val	Disease	Train	Val	Test
Normal (N)	2261 (46.2%)	482 (46.1%)	AMD (A)	326 (6.7%)	67 (6.4%)	66 (6.4%)
Diabetic Retinopathy (D)	1267 (25.9%)	273 (26.1%)	Hypertensive Retinopathy (H)	135 (2.8%)	28 (2.7%)	30 (2.9%)
Glaucoma (G)	227 (4.6%)	49 (4.7%)	Pathological Myopia (M)	187 (3.8%)	40 (3.8%)	41 (4.0%)
Cataract (C)	225 (4.6%)	46 (4.4%)	Other (O)	719 (14.7%)	155 (14.8%)	153 (14.9%)

Table 2: Different degrees of multi-labels existing in entire cleaned dataset.

Diseases per Image	Train	Val	Test
1 disease	4463 (91.2%)	953 (91.2%)	936 (90.9%)
2 diseases	413 (8.4%)	89 (8.5%)	94 (9.1%)
3 diseases	18 (0.4%)	3 (0.3%)	0 (0.0%)
4 diseases	1 (0.0%)	0 (0.0%)	0 (0.0%)
TOTAL	4895 (100.0%)	1045 (100.0%)	1030 (100.0%)

The 15% test set containing 1030 images was reserved as unseen data for evaluation. To further test generalization, an external CFD dataset such as BRSET [11] will be used to evaluate on the final model. This may reveal generalization challenges as different datasets come from different demographics and imaging equipment. Another challenge the current dataset inherits is class imbalance, which means the model may develop biases towards majority classes and poorly detect rarer conditions. Additional data may be needed for future training to address these limitations.

Baseline Model

ResNet-50 is a common backbone in ocular disease classification research, often achieving high accuracy [12]. Thus, ResNet-50 pre-trained on ImageNet with a simple custom head for 8 classes is used as the baseline. The baseline model was trained with the backbone frozen, a batch size of 32, a learning rate of 0.001, the Adam optimizer, binary cross-entropy loss, and no hyperparameter tuning. The model was saved at epoch 19, when the validation error was minimal.

The baseline achieved a test error of 11.59% and a macro F1 of 0.2954. This discrepancy reveals the challenge posed by class imbalance. Since the model is trained on a larger proportion of common classes (like N and D) and the test set contains a similarly larger proportion of common classes, the strong performance in common classes dominates the test error, making it artificially low. However, macro F1 treats each class equally, meaning the poor performance in underrepresented diseases is reflected in the low score.

The confusion matrix reveals that the baseline model is biased towards being conservative. It tends to predict the absence of a disease across disease classes, evident from the higher number of false negatives compared to false positives. Overall, these results show that the baseline model is unsuitable for clinical usage.

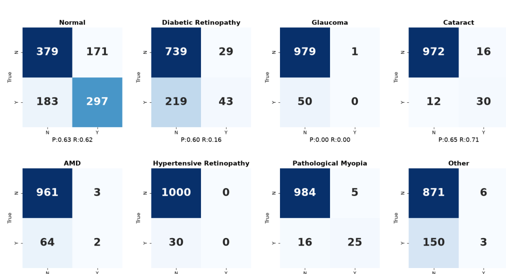

Figure 2: Baseline model confusion matrix.

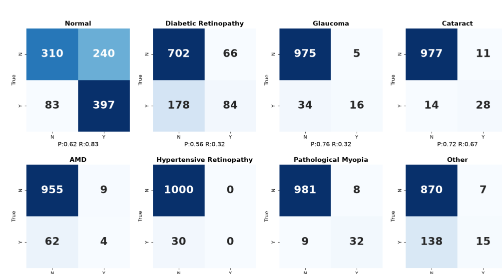

Figure 3: OcuDetect model confusion matrix.

Primary Model

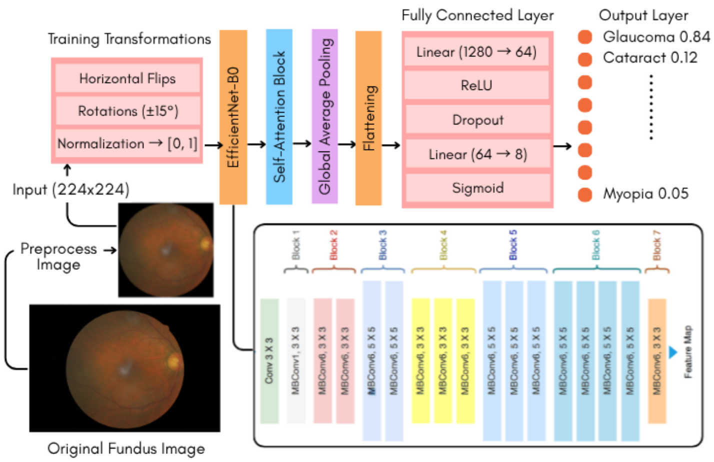

Figure 4: OcuDetect Version 1 schematic diagram.

The current best OcuDetect model (Figure 4) employs the EfficientNet-B0 pre-trained on ImageNet as a backbone with multiple convolutional layers. Self-attention can benefit multi-label classification, as some ocular diseases are characterized by their non-local dependencies. For example, the relative spatial arrangement of patterns like microaneurysms and exudates can indicate the diabetic retinopathy class [13]. ViTs requires large datasets to generalize [14], making them ill-suited for the small CFP datasets available; thus, a self-attention block is added to the CNN backbone to test its feasibility. A Global Average Pooling layer is applied after the self-attention block, followed by fully connected layers with ReLU activation. The output layer uses a sigmoid activation function for multi-label classification. OcuDetect was trained with the backbone frozen until after 5 epochs, a batch size of 32, a learning rate of 0.001, the Adam optimizer, and binary cross-entropy loss. The model has 5,010,628 parameters and 1,003,080 (20.0%) trainable paramters. The model was saved at epoch 16, when the validation error was minimal. Past this point, the model started to overfit (Figure5).

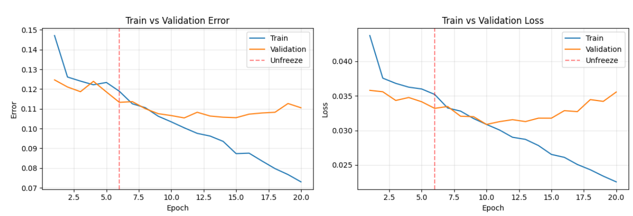

Figure 5: Learning curves for OcuDetect.

To reduce overfitting, data augmentation, dropout, and a high compression layer in the classifier head were added. However, overfitting still remains a challenge, as shown in the learning curve. One suspected reason is the small dataset size paired with the data-hungry self-attention feature. Finding larger datasets may help and better reveal the capabilities of the model architecture.

The confusion matrix (Figure3) reveals that OcuDetect shows better precision and recall in certain classes, while the baseline performs better in others.

The current best OcuDetect model achieved a test error of 10.85%, which is a slight improvement above baseline, and a macro F1 of 0.4154. Compared to baseline, the F1 score improved across all classes other than H. This indicates that while the overall error reduction is slight, the performance is substantially more balanced than baseline. This improvement warrants further investigation into OcuDetect's potential, particularly for improving the detection of rarer and underrepresented diseases.

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